higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MECHANOTECHNICS N6

3 APRIL 2019

This marking guideline consists of 11 pages.

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QUESTION 1

$$W = mg = 50 \times 9{,}81 = 490{,}5 \quad N$$

$$\Sigma CWM = \Sigma ACWM$$

$$(490{,}5 \times 500) + [O\,,32\,R(150 - 50)] = R \quad \times 150$$

$$245250 = 150\,R - 32\,R$$

$$R = 2078{,}39 \quad N \qquad (4)$$

1.2 $$T = (\mu R)\,r$$
$$= (0{,}32 \times 2078{,}39)\,0{,}15$$

$$= 99{,}763 \quad Nm \qquad (2)$$

1.3 $$P = \frac{2\pi NT}{60}$$
$$\frac{2\pi(1440)(99{,}763)}{60}$$

$$P \quad = 15{,}044\,Kw \qquad (2)$$

[8]

QUESTION 2

2.1 $$P = \frac{2\pi NT}{60}$$

$$25 \times 10^3 = \frac{2\pi(950)\,T}{60}$$

$$T = 251{,}3\,Nm$$

$$T = \mu(F_c - S)\,R \times n$$

$$251{,}3 = 0{,}3(F_c - S)\,0{,}2 \times 4$$

$$F_c - S = 1047{,}072\,N$$

$$F_c - S = m\left[w_1^2 - w_2^2\right]r \qquad \text{but} \qquad w_1 = \frac{2\pi N}{60} = \frac{2\pi(950)}{60}$$

$$= m\left(99{,}484^2 - 69{,}639^2\right)0{,}15 \qquad = 99{,}484\,rad/s$$

$$m = 1{,}383\,kg \qquad w_2 = \frac{2\pi N}{60} = \frac{2\pi(0{,}7 \times 950)}{60}$$

(8)

$= 69,639\ rad/s$

2.2 $S = m w_{2^2} r$

¿$(1,383)(69,639)^2\ 0,15$

$= 1006\ N$

$F_c = m w_1{}^2 r$

$= (1,383)(99,639)^2(0,15)$

$= 2059,549\ N$

$S = \frac{F}{x}$

$160 = \frac{F}{3}$

$F = 160 \times 3 = 480\ N$

$New\ \ spring\ \ force = 1006 + 480 = 1486\ N$

$F_c - S = 2059,549 - 1486 = 573,54\ \ 9$

$T = \mu(F_c - S)Rn = 0,3(573,549)0,2 \times 4$

$= 137,652\ Nm$ (7)

[15]

QUESTION 3

3.1 $F_t = \frac{T_A \times 2}{PCD}$

$= \frac{543 \times 2}{0,12}$

$= 9050\ \ N$ (1)

3.2 $F_R = F_t \tan \varnothing$

$= 9050 \tan 20$

$= 3293,931\ \ N$ (1)

3.3 $T = (T_1 - T_2)R$

$296 = (3,5T_2 - T_2)0,15$ (3)

$T_2 = 789{,}333 \ N$

$T_1 = 2762{,}667 \ N$

3.4 $\Sigma CWM = \Sigma ACWM$

$(810 \times 300) + (9050 \times 600) + [(2762{,}667 + 789{,}333)\cos 30°](1050) = (R_R \times 900)$

$R_R = 9892{,}142 \ N$

$(R_L \times 900) + [(2762{,}667 + 789{,}333)\cos 30°]150 = (9050 \times 300) + (810 \times 600)$

$R_L = 3043{,}97 \ N$

$990{,}297 = (3{,}514\,T_2 - T_2)0{,}2$ (4)

3.5 $\Sigma CWM = \Sigma ACWM$

$(3293{,}93 \times 600) + [(2762{,}667 + 789{,}333)\sin 30°]1050 = R_R \times 900$

$R_R = 4267{,}954 \ N$

$\Sigma CWM = \Sigma ACWM$

$(R_L \times 900) + [(2762{,}667 + 789{,}333)\sin 30°]150 = 3293{,}913 \times 300$

$R_L = 801{,}971 \ N$ (4)

[13]

QUESTION 4

4.1 $m = \rho v$

$= (7800)[\frac{\pi}{4}(0,12)^2 \times 1,2]$

$= 105,859\,kg$

$I = \frac{1}{2}mr^2$

$= \frac{(105,859)(0,06)^2}{2}$

$= 0,191\,kg.m^2$ (4)

4.2 $m = \rho v$

$= (7800)[\frac{\pi}{4}((0,24)^2 - (0,12)^2) \times 0,12]$

$= 31,758\,kg$

$I = \frac{1}{2}m(R^2 + r^2)$

$= \frac{1}{2}(31,758)[(0,12)^2 + (0,06)^2]$

$= 0,286\,kg.m^2$ (4)

4.3 $m = \rho v$

$= (7800)[\frac{\pi}{4}(0,83)^2 - (0,65)^2]$

$= 163,199\,kg$

$I = \frac{1}{2}m(R^2 + r^2)$

$= \frac{1}{2}(163,199)[(0,415)^2 + (0,325)^2]$

(4)

$= 22,672\ kg\,.\,m^2$

4.4 $m = \rho v$

$= (7800)(1300 \times 10^{-6} \times 8 \times 0,205)$

$= 16,63\ kg$

$I = \frac{ml^2}{12} + m r^2$

$= \frac{(16,63)(0,205)^2}{12} + (16,63)(0,2225)^2$

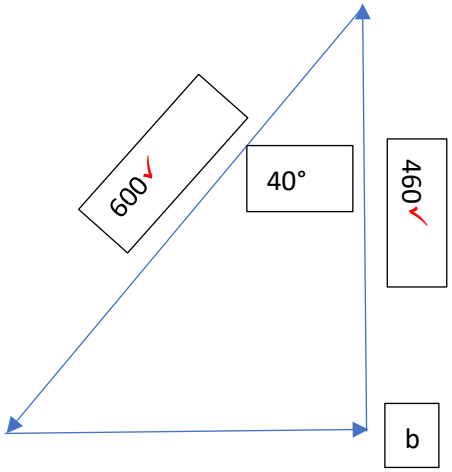
$= 0,882\ kg\,.\,m^2$ (4)

4.5 $I_T = 24,031\ kg/m^2$ (1)
[17]

QUESTION 5

5.1

	Mass(kg)	r(mm)	mr(kg.mm)	l ¿	$mr\,l$ ¿
A	4	280	1120	+(0,35+x)	+392+1120 x
B	5	220	1100	+0,35	+385
C	m	200	200m	0	0
D	3	250	1500	-0,4	-600



$Angle\ \ between\ \ A \wedge D = 40°$

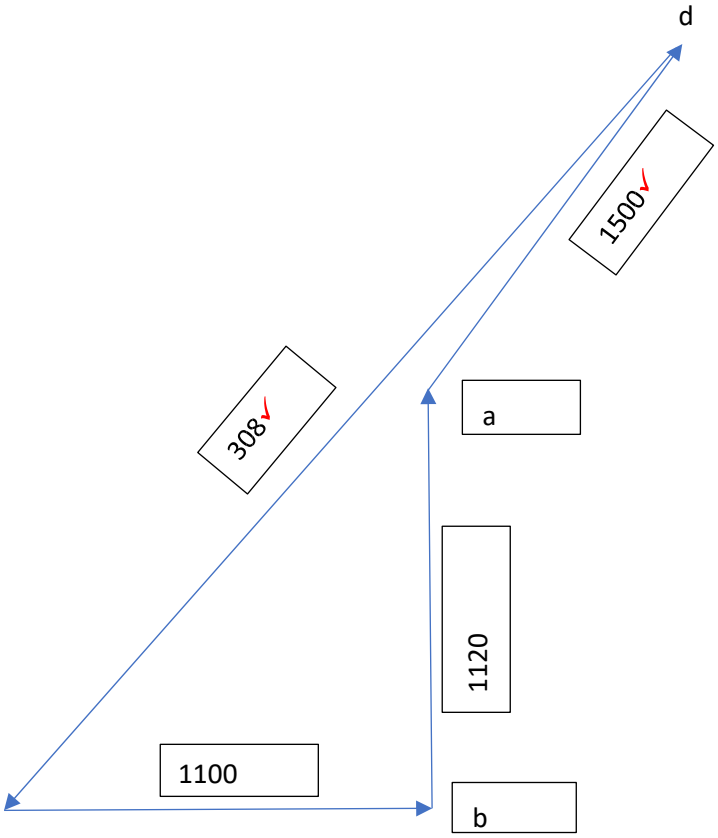
$392 + 1120\,x = 460$

$x = 60{,}7 \times 10^{-3}\,m$

$x = 60{,}7\,mm$

(11)

5.2



$308 = 200\,m$

$m = 1{,}54\,kg$

$Angle\ \ between\ \ B\ \wedge\ C = 137\,°$ (7)
[18]

QUESTION 6

6.1 $$tan\theta = \frac{lead}{\pi \times PCD}$$

$$tan\theta = \frac{10 \times 3}{\pi(60)}$$

$$\theta = 9{,}043\,°$$

$$\eta_w = \frac{tan\theta}{\tan(\theta + \varnothing)} \times 100\,\%$$

$$70\,\% = \frac{\tan(9{,}043)}{\tan(9{,}043 + \varnothing)} \times 100\,\%$$

$$\tan(9{,}043 + \varnothing) = 0{,}227$$

$$9{,}043 + \varnothing = \tan^{-1}(0{,}227)$$

$$9{,}043 + \varnothing = 12{,}809$$

$$\varnothing = 3{,}766\,°$$

$$\mu = \tan\varnothing$$
$$\mu = \tan 3{,}776$$

$\mu = 0{,}066$ (8)

6.2 $P = \frac{2\pi NT}{60}$

$20 \times 10^3 = \frac{2\pi(600)T}{60}$

$T = 318{,}31\ Nm$ (2)

6.3 $P = Fv$

$F = \frac{20 \times 10^3}{1{,}885}$

$F = 10{,}61\ kN$

$v = \frac{\pi \times PCD \times N}{60}$

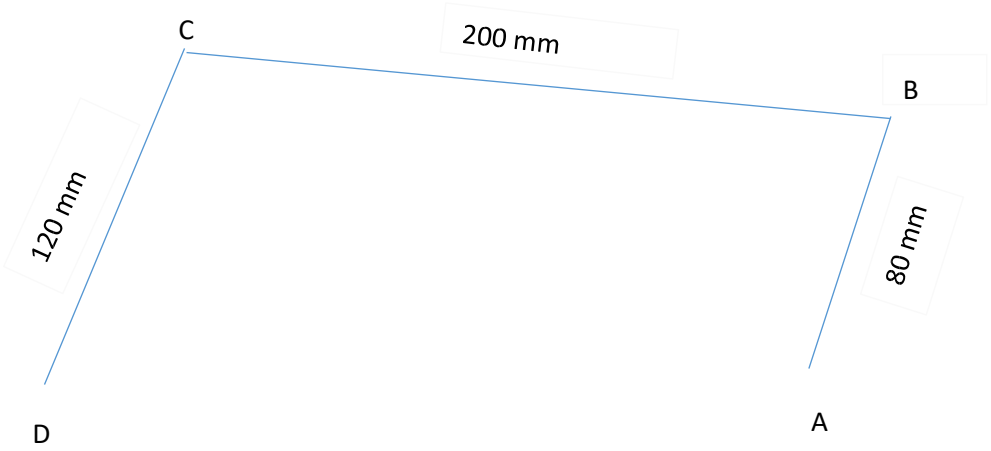
$¿ \frac{\pi \times 0{,}06 \times 600}{60}$

$= 1{,}885\ m/s$ (2)

[12]

QUESTION 7

7.1



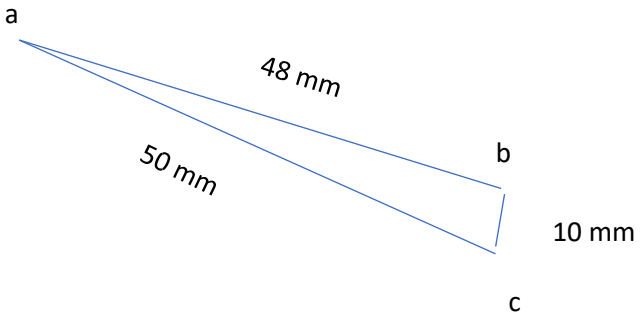
(3)

7.2 $v_B = wr$

$= 10 \times 0{,}08$

$= 0{,}8\ m/s$ (1)

7.3



(3)

7.4 $a_{AB} = w^2 AB$

$= (10)^2 (0,08)$

$= 8\, m/s^2$

$a_{BC} = \frac{v^2}{BC}$

$= \frac{8 \div 60}{0,2}$

$= 0,667\, m/s^2$

$a_{CD} = \frac{v^2}{CD}$

$= \frac{50 \div 60}{0,12}$

$= 6,944\, m/s^2$

(3)

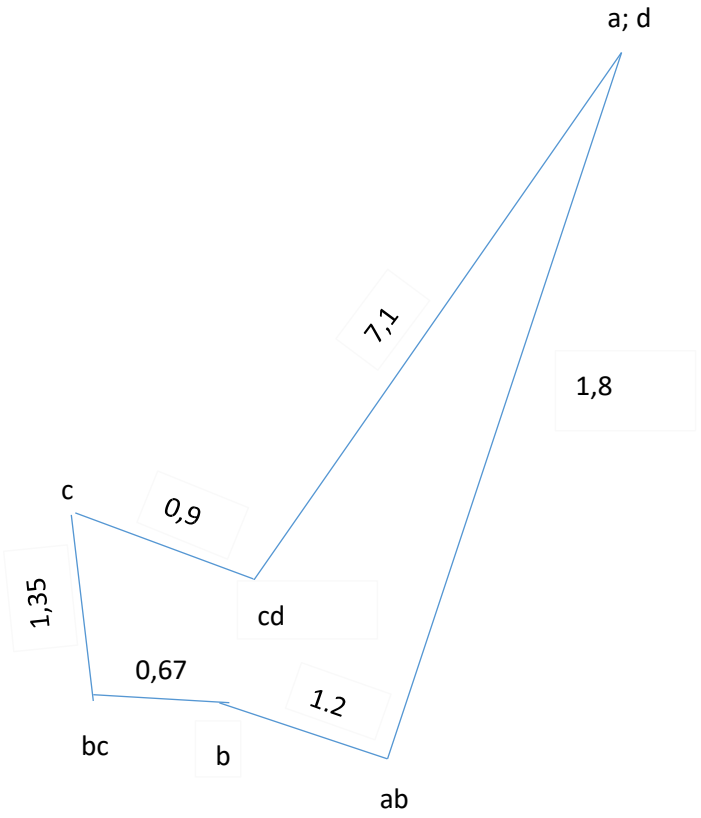
7.5 $a_{AB} = \propto AB$

$= (15)(0,08)$

$= 1,2\, m/s^2$ (1)

7.6

(6)



[17]

TOTAL: 100

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